

QA Automation Interview Study Guide (Full)

Java, Selenium, SQL, and QA Design Q&A; with Code Snippets

Mock Interview Q&A;

Java: == vs .equals()

Interviewer: What's the difference between == and .equals() in Java? Answer: == compares memory references. .equals() compares values inside objects. For Strings, two objects with same text will return false with == but true with .equals().

Java: Framework scalability

Interviewer: How would you make your automation framework scalable and maintainable? Answer: Use Page Object Model to separate locators from test logic, utility classes for common actions, externalize configuration in properties files, and structure project with Maven + TestNG for CI/CD integration.

Selenium: Synchronization issues

Interviewer: How do you handle synchronization issues? Answer: Use explicit waits with WebDriverWait and ExpectedConditions. This gives precise control compared to implicit waits.

Selenium: Dynamic element IDs

Interviewer: Suppose an element ID changes every time. How do you handle it? Answer: Use stable locators like name or text, or a dynamic XPath with contains/starts-with.

Selenium: Multi-browser execution

Interviewer: How do you run Selenium tests on multiple browsers? Answer: Parameterize browser type in config or TestNG, initialize driver accordingly, and use Selenium Grid or cloud providers for parallel execution.

SQL: Validate user registration

Interviewer: How would you validate that a new user is saved in DB? Answer: After Selenium registration, connect to DB via JDBC and query Users table with username/email. Validate record fields like registration date and status.

SQL: Find duplicates

Interviewer: How do you find duplicate records? Answer: Use GROUP BY and HAVING COUNT(*) > 1. Example: SELECT email, COUNT(*) FROM Users GROUP BY email HAVING COUNT(*) > 1;

QA: Login test cases

Interviewer: Walk me through test cases for a login page. Answer: Positive: valid username/password. Negative: invalid password, invalid username, empty fields. Also test boundary values and SQL injection attempts.

QA: Choosing automation vs manual

Interviewer: How do you decide which cases to automate? Answer: Automate repetitive and regression-heavy tests. Keep exploratory or volatile tests manual.

QA: Smoke vs sanity vs regression

Interviewer: Explain the difference. Answer: Smoke = basic stability check. Sanity = focused validation after a bug fix. Regression = full suite to confirm old features still work.

End-to-End: E-commerce order

Interviewer: How would you automate testing order placement? Answer: Use Selenium to place order, validate confirmation on UI, then query Orders table with JDBC to ensure DB reflects correct items, prices, and status.